

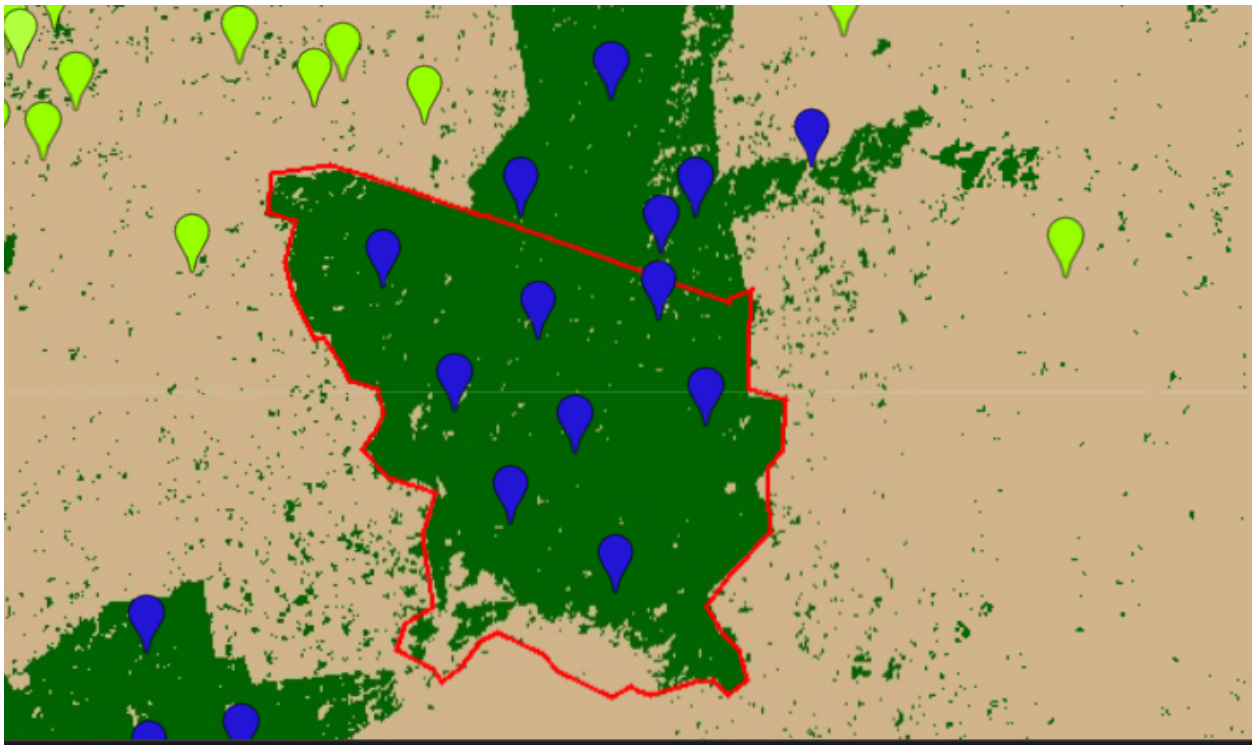
Forest carbon MRV prototype

Kakum National Park, Ghana | Sentinel-2 & Google Earth Engine | 2023

A satellite-based estimate of how much forest stands inside Kakum National Park, and how much carbon it holds. Built with free Sentinel-2 imagery, following the measurement logic of a Verra REDD+ project, and reported with a real accuracy figure and an honest uncertainty range.

Headline results (2023)

Metric	Value
Park boundary (WDPA “Kakum”)	21,168 ha
Forest area	18,800 ha (≈89% of the park)
Classification accuracy	95.5% (n = 22 test points)
Above-ground carbon density	146 tC/ha (534 tCO ₂ e/ha)
Carbon stock, above-ground	10.0 million tCO ₂ e
Carbon stock, incl. below-ground	12.5 million tCO ₂ e
Combined uncertainty	±61.5%
Conservative estimate (lower bound)	4.8 million tCO ₂ e



Forest (dark green) and non-forest (tan) inside Kakum, clipped to the official WDPA boundary (red). Blue pins are forest training points, green pins non-forest.

What this is, and what it isn't

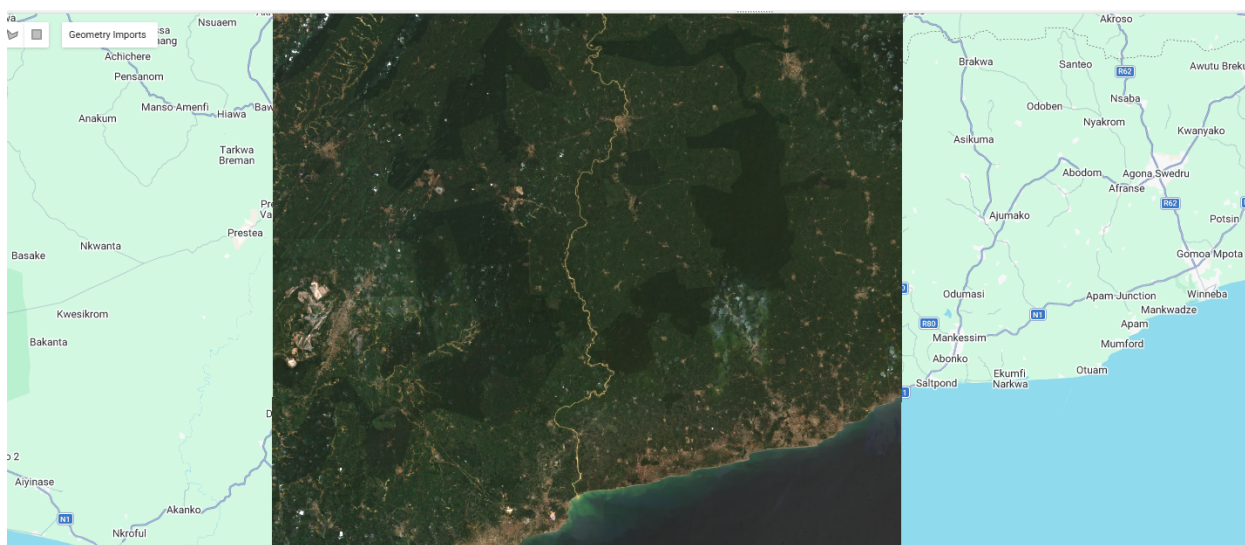
At its simplest, this takes satellite imagery, maps the forest, turns that forest into a carbon figure using IPCC default values, and reports the answer with an uncertainty range and a conservative floor.

What it isn't: a carbon credit, or a project registration. One structural reason is worth being upfront about. Since late 2023, Verra's consolidated REDD+ methodology (VM0048) no longer lets the project developer set the baseline. Verra sets it from jurisdictional deforestation data. So a real registration would swap out part of what I built for Verra's own baseline layer. What I've reproduced is the part a developer actually builds and has to defend, the activity data and the emission factors, and I'm clear about the part that now lives outside their hands.

Study area and boundary

Kakum sits in Ghana's Central Region, about 30 km north of Cape Coast, in the Upper Guinean forest belt.

Everything here is clipped to the official park boundary from the World Database on Protected Areas (WDPA), and the boundary is worth a careful sentence. The WDPA polygon named "Kakum" covers 21,168 ha. The number you'll usually see quoted for Kakum is 375 km² (37,500 ha), but that figure describes the wider complex, the national park plus the Assin Attandanso Resource Reserve next to it. This estimate is for the national park on its own. Which boundary you pick changes the answer, so the report states the one it used rather than leaving it ambiguous.



True-colour Sentinel-2 composite (2023) over the study area. The dark-green forest mass stands out against the lighter farmland around it and the coast to the south. That contrast is what the classifier turns into a forest map.

Data

- **Imagery:** Sentinel-2 surface reflectance, harmonized
- **Period:** 2023 calendar year
- **Cloud handling:** scenes below 20% cloud cover, combined into a median composite (10 scenes passed)
- **Bands used:** blue, green, red, near-infrared, and two shortwave-infrared

Method

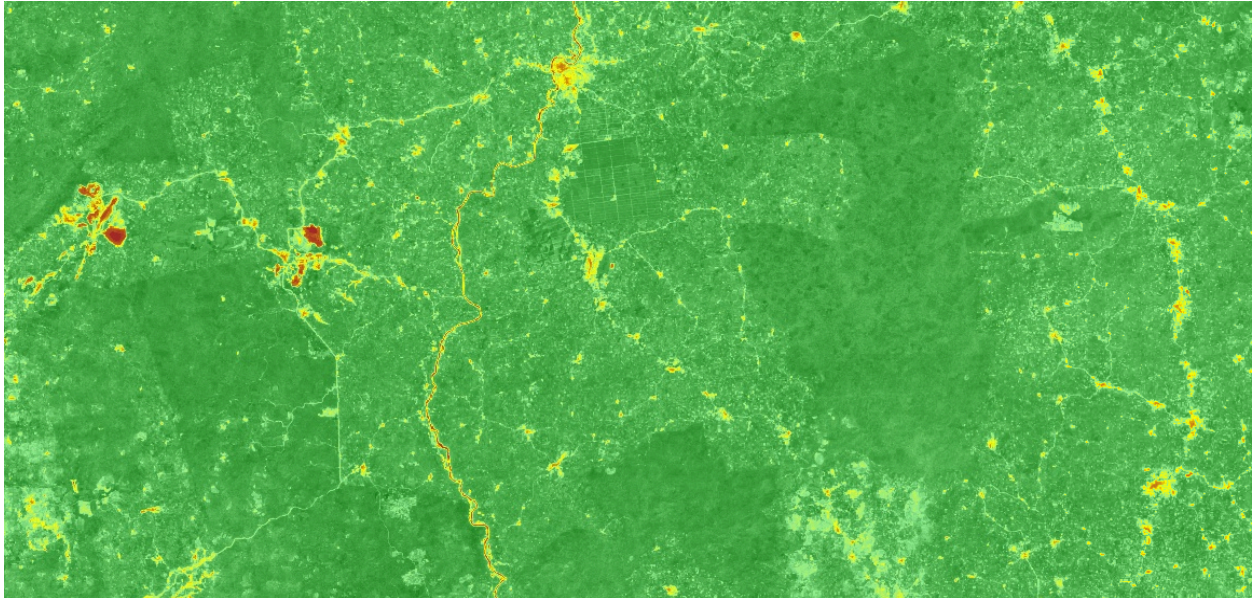
Why not just use NDVI

My first instinct was the obvious one: threshold NDVI, the standard greenness index, and call everything above the line forest. On this landscape that falls apart, and the numbers show exactly why. Measured on the composite:

- Forest $\approx$ **0.76**
- Neighbouring farmland $\approx$ **0.72**

— Town $\approx$ 0.31

NDVI does a clean job of splitting vegetation from bare ground and buildings, which is why the town drops out at 0.31. But forest and farmland sit only about 0.04 apart, and that gap is inside the ordinary pixel-to-pixel noise. In humid West Africa, dense farms and tree crops are almost as green as forest, so greenness on its own can't draw the forest line.



NDVI over the study area. Forest and farmland are nearly indistinguishable; only towns and roads (yellow) separate cleanly. This is why a single greenness threshold won't work here.

The classifier

So I switched to a supervised Random Forest (50 trees), trained on all six bands. The shortwave-infrared bands are the ones doing the real work: they carry moisture and canopy-structure signal that tells forest apart from farmland even when the two look identical in greenness. Training data was 40 forest points and 40 non-forest points, placed by hand across the scene.

One early run got a tree-crop plantation wrong, painting it as forest. That's a fair mistake for the model to make, since a mature plantation looks a lot like natural forest to a satellite. Adding a handful of non-forest points inside the plantation fixed it. I'm calling that out rather than quietly correcting it, because plantation-counted-as-forest is one of the classic ways REDD+ projects overstate carbon, and knowing to watch for it is half the skill.

How accurate is it

I split the training points 70/30 into a set the model learns from and a set it never sees, with a fixed random seed so the split reproduces. Then I scored the model on the held-back points:

- **Overall accuracy: 95.5%** on 22 test points
- Confusion matrix: 11 non-forest right (1 called forest), 10 forest right (none missed)
- The one error leans toward slightly over-mapping forest

Honest caveat: the test points came from the same clicking session as the training points, so this figure is a little flattering compared with a properly independent reference sample. It's a solid number, not a perfect one, and it's more believable for not being 100%.

Forest area

Converting the map to hectares is a per-pixel area sum inside the boundary:

- Forest: **18,800 ha**
- Non-forest: 2,278 ha
- Total: 21,076 ha

That total lands within 0.4% of the polygon's own geometric area (21,168 ha), which is a nice independent check that the classification actually covers the whole park with no holes. Forest works out to roughly 89% of the park, which is what you'd hope to see inside a protected area that's still largely intact.

From hectares to carbon

The conversion uses IPCC Tier 1 defaults for African tropical rainforest:

- Above-ground biomass: 310 t dry matter/ha (wide plausible range, roughly 130–510)
- Carbon fraction: 0.47
- Carbon to CO₂: $\times 44/12$
- Root-to-shoot ratio: 0.24 (to add below-ground biomass)

That gives an above-ground density of 146 tC/ha (534 tCO₂e/ha), and applied to the mapped forest:

- Above-ground stock: **10.0 million tCO₂e**
- Including roots: **12.5 million tCO₂e** (662 tCO₂e/ha)

The per-hectare density falls inside the range reported for tropical rainforest, which is the quick gut-check that the arithmetic isn't off by a factor of ten somewhere.

One distinction matters if anyone reads this as a carbon claim. This is a **stock**, the carbon standing in the forest right now. It isn't an **emission**. Emissions are a change over time, the carbon released when forest is cleared, and measuring them needs a second date to see what disappeared. The per-hectare figure here is the emission factor you'd multiply against forest loss once you have that second date.

Putting an error bar on it

A lone carbon number pretends to a precision it doesn't have, so I combined the two big sources of error:

- Area uncertainty (from the classification accuracy): $\approx$ **4.5%**
- Biomass uncertainty (from the IPCC default range): $\approx$ **61%**

These don't just add. Independent uncertainties combine in quadrature, the square root of the sum of squares, which comes to **61.5%**. And that result is the interesting part: the combined figure is barely above the biomass figure on its own. Once you square each source, the big one swallows the small one, and the map's accuracy ends up contributing almost nothing to the total. The biomass default carries the whole thing.

On a central estimate of 12.5 million tCO₂e, that's a margin of about ± 7.65 million, so a range of **4.8 to 20.1 million tCO₂e**. Carbon accounting says you should never overstate, so the number I'd actually stand behind is the bottom of that range: **4.8 million tCO₂e**.

Limitations

- **One date.** This is a 2023 stock, not an emissions estimate. Emissions need at least two dates.
- **Tier 1 biomass.** A continent-average biomass value dropped onto one specific forest is, by a wide margin, the biggest source of error.
- **Carbon pools.** Only live biomass (above and below ground) is counted. Dead wood, litter, and soil carbon are left out, and in forests those can be large.
- **Validation sampling.** Test and training points share a source, so the accuracy runs slightly optimistic. A rigorous design uses independent reference points.

- **Simplified uncertainty.** Area error is proxied by $(1 - \text{accuracy})$ rather than propagated through the full confusion matrix, and the conservative deduction follows the VCS principle, not its exact tables.
- **Resolution.** Narrow rivers blur out at 10 m, because each pixel mixes water with the dense vegetation on the banks. Galamsey mining turbidity is there in the rivers, but it's a small part of those pixels.

What I'd do next

- **Replace the Tier 1 biomass default with a site-specific layer** from GEDI or radar. The error budget makes this the single highest-value move, ahead of any tweak to the classifier, because biomass is what dominates the uncertainty.
- **Add a second image date** (say 2018) to turn the stock estimate into a measurement of forest loss and the emissions that go with it, which is the number a REDD+ project ultimately reports.
- **Collect independent reference points** to firm up the accuracy figure.

Reproducibility

The whole analysis is one Google Earth Engine script, run start to finish with a fixed random seed, so the reported numbers come back the same on a re-run. The only manual input is the training points (two geometry layers, forest and nonforest), described in the script header.

Data sources

- Sentinel-2 surface reflectance (harmonized), via Google Earth Engine
- WDPA protected-area boundaries, via Google Earth Engine
- IPCC 2006 Guidelines and 2019 Refinement: default biomass and carbon values
- Reference methodology: Verra VM0048 (consolidated REDD+)